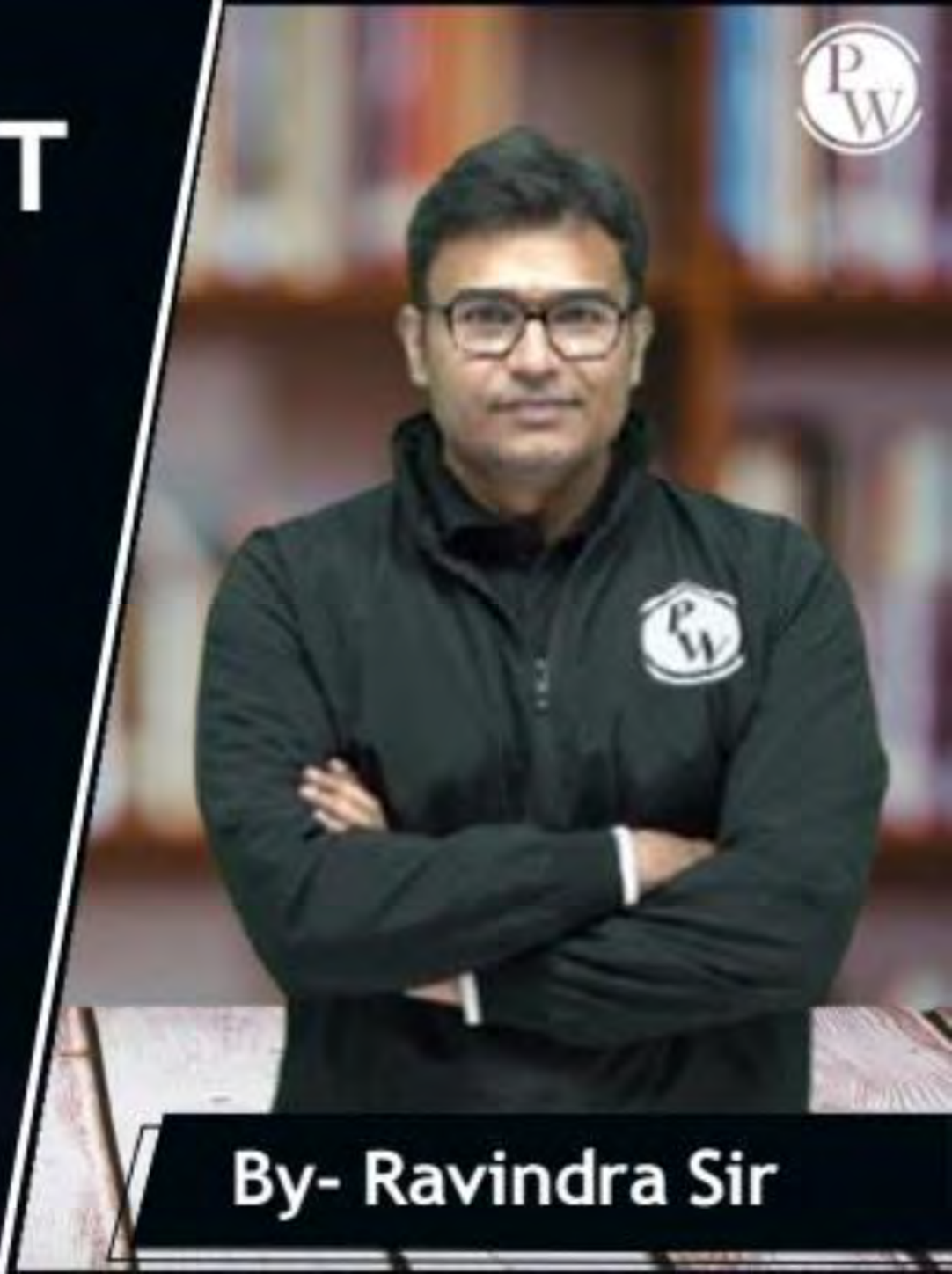
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO -OSI Stack

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

ISO - OSI

A
P
S
T
N
D ✓
P ✓

Topics to be Covered



- Topic
- Topic
- Topic

DLL

Inspiring Stories : Moji Riba



Background: Born 1911 in a remote village in Arunachal Pradesh.

Struggles: His people were ignored and kept cut off from the rest of India.

Education: Went to a christian school, rare for his region.

Achievements: Guided American soldiers in wartime (spoke English), helped bring modern tools and awareness to his region. Later honored by the government with awards and a postage stamp.

Impact: Proved even the most remote people could change history.

Inspiring Stories : Bhima Nayak



Background: Born in 1840 in Barwani, Madhya Pradesh, from the Bhil tribe.

Struggles: Saw his people forced to work without pay and pay heavy taxes.

Education: Learned from his tribe and the hills.

Achievements: Joined the 1857 fight, fought the British, later sent to jail in Andaman.

Impact: Bhil people still remember him as a brave leader who fought for them.

Inspiring Stories : Kiran Bala Bora



Background: Born 1904 in a small village in Assam.

Struggles: Faced people who didn't want girls to study.

Education: Studied only till Class 3.

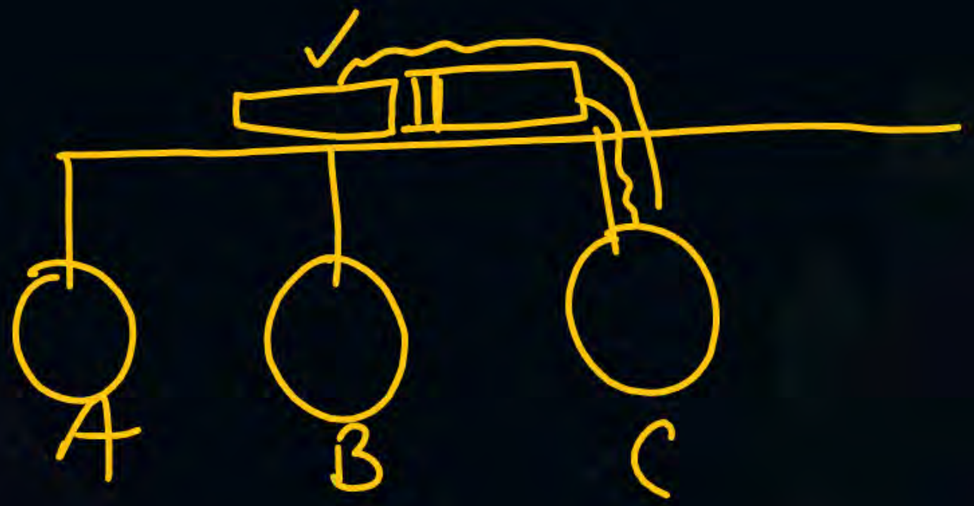
Achievements: Joined protests in the 1930s and pushed women to lead.

Impact: Proved that courage matters more than schooling.

DLL :-

1. Flow control → GBN
2. Error control → CRC
3. Framing ✓
4. Physical add
5. Access control (CSMA/CD) ✓ → ethernet → CSMA/CD


DA
C
S
FD


A
B
C

Framing

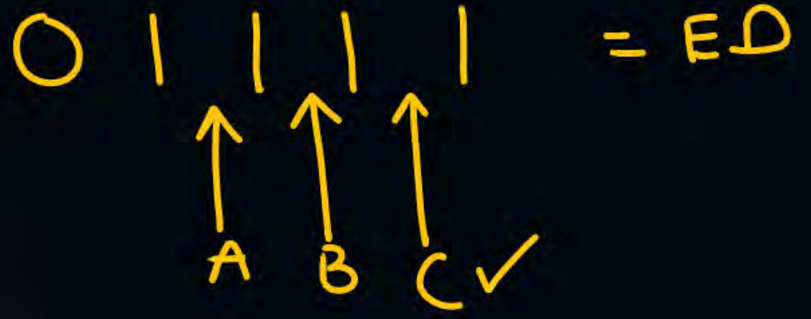
- Fixed length
  - X
  - 1000
  - E
  - S
- Variable length
  - Length
    - 500
    - Ethernet
  - End delimiter
    - Character
      - Character Stuffing
    - Pattern
      - Bit Stuffing





This method is obsolete.





Bit Stuffing

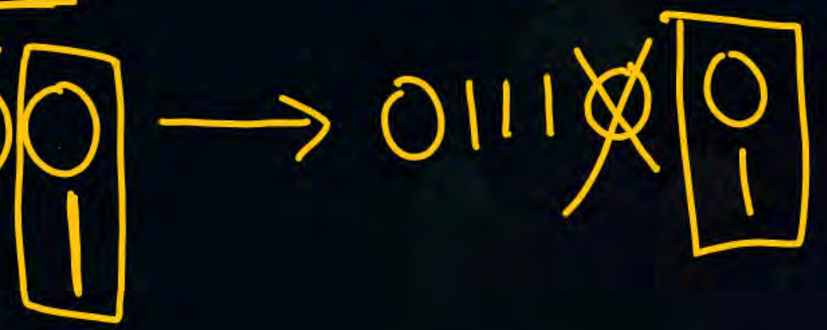
ED: 01111

Sender

Data: 01110 → Rec 0111 0 1

Data: 01110 → 0111 0 1

01110 → 0111 0 0



ED: 01111

Sender:-

After every 0111 add a zero

Rec:-

After every 0111 remove a zero

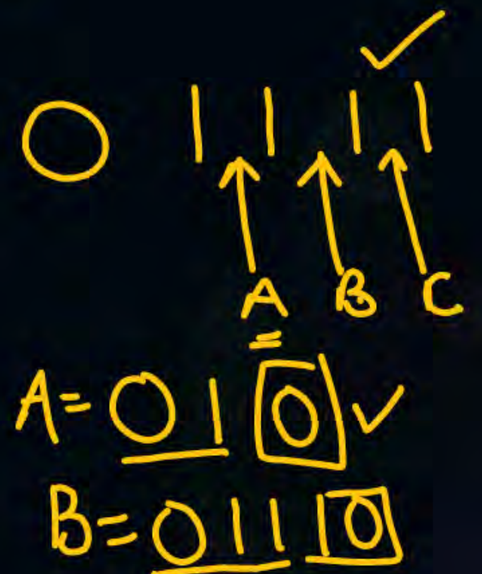
Data:- ED:- 01111

Send:- 0 1 1 1 0 0 1 1 1 0 1 0 0

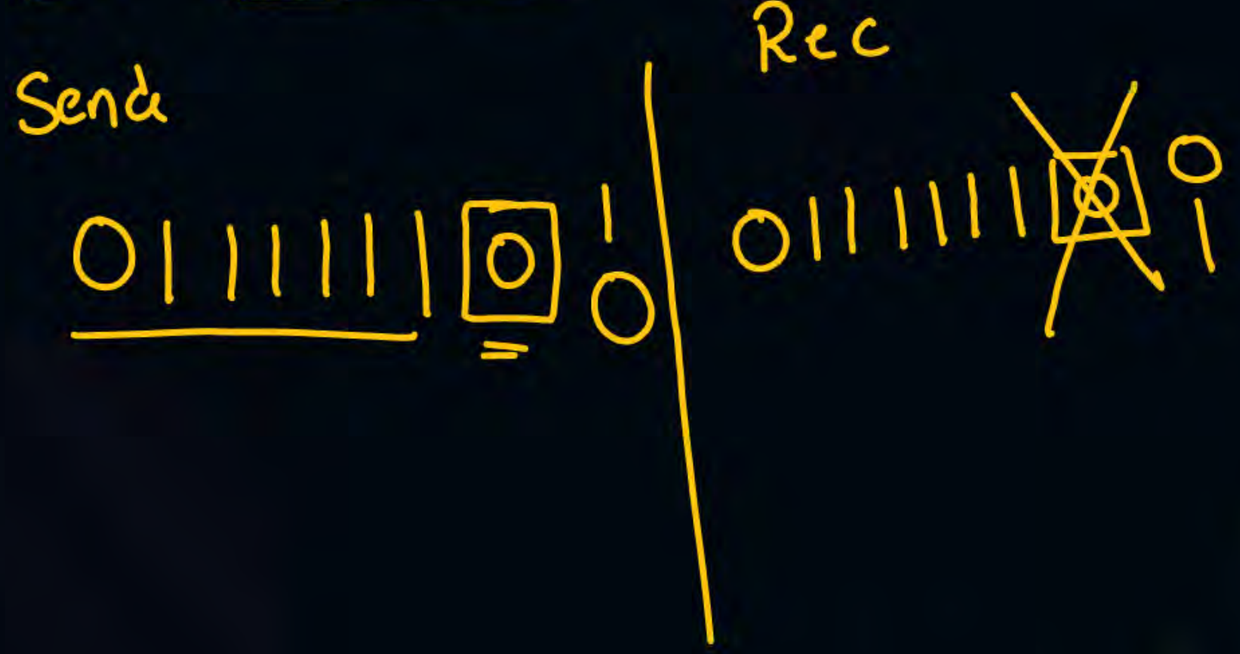
Rec:- 0 1 1 1 ~~0~~ 0 1 1 1 ~~0~~ 0 0

01111 = ED

0111 0



ED :- 01111111



ED: 10000000

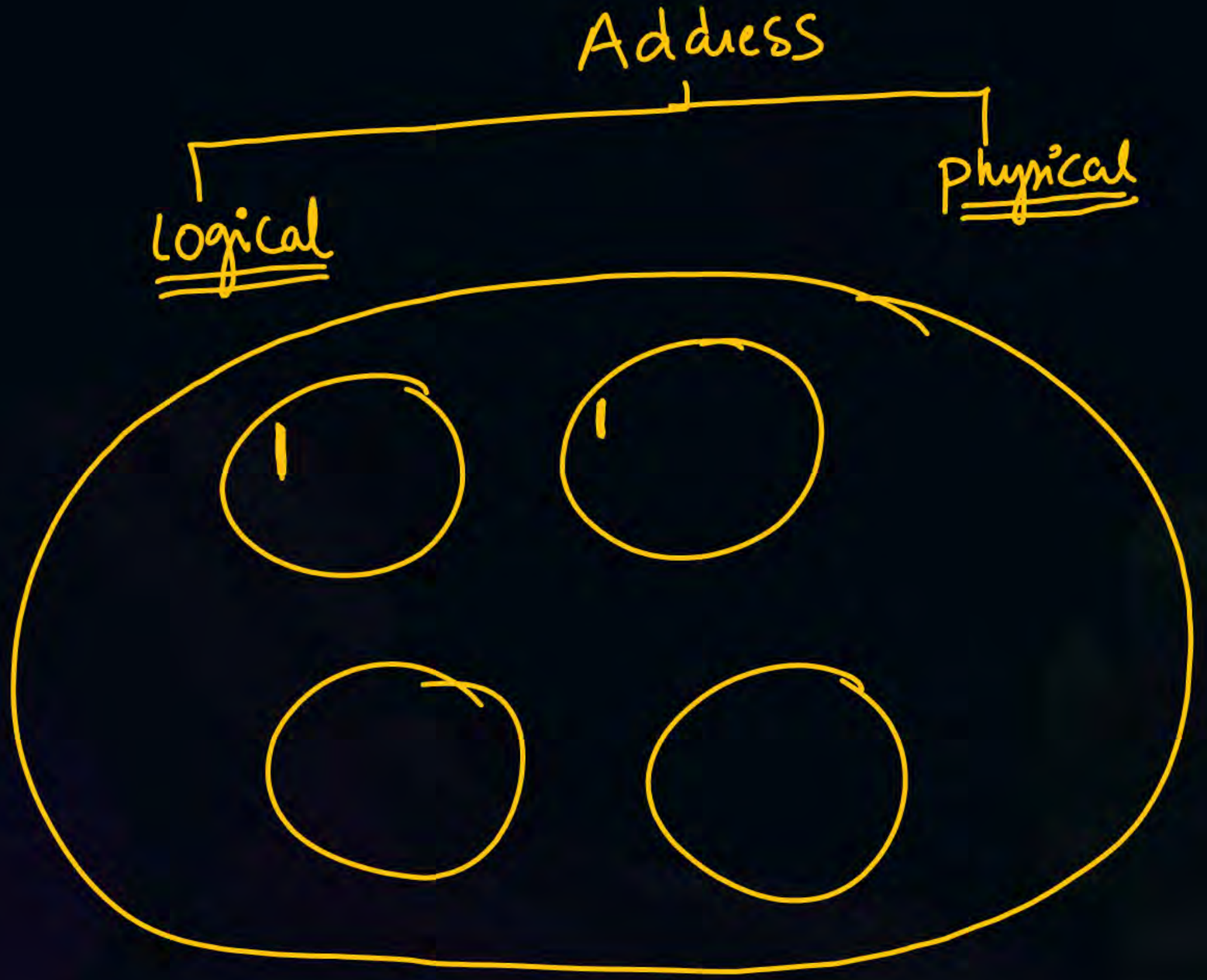
Sen: Rec:



Bit stuffing

Physical addressing :-

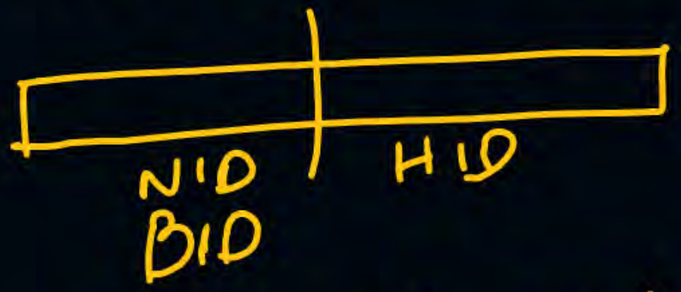
DLL deals with physical addressing


Address
Logical
Physical

Concepts { Logical add :- Any number that is unique globally.
Physical add :- Any number that is unique within a LAN.

IP add and MAC add
Reality

IP – 32 bit number – S/w number – globally unique



MAC – 48 bit number – H/w number (ROM of NIC) – globally unique



LA → globally unique

IP, MAC → globally unique

1) Can IP add be used as logical add?

A) Yes, we are using it because IP add contains NID, HID which will help in routing

2) Can MAC add be used as LA?

A) Yes, but we are not using it, because MAC doesn't have NID, HID and so not useful for routing

3) Can IP add be used as physical add?

A) Yes, but we are not using it. Because it is not permanent.

3) Can MAC add be used as PA?

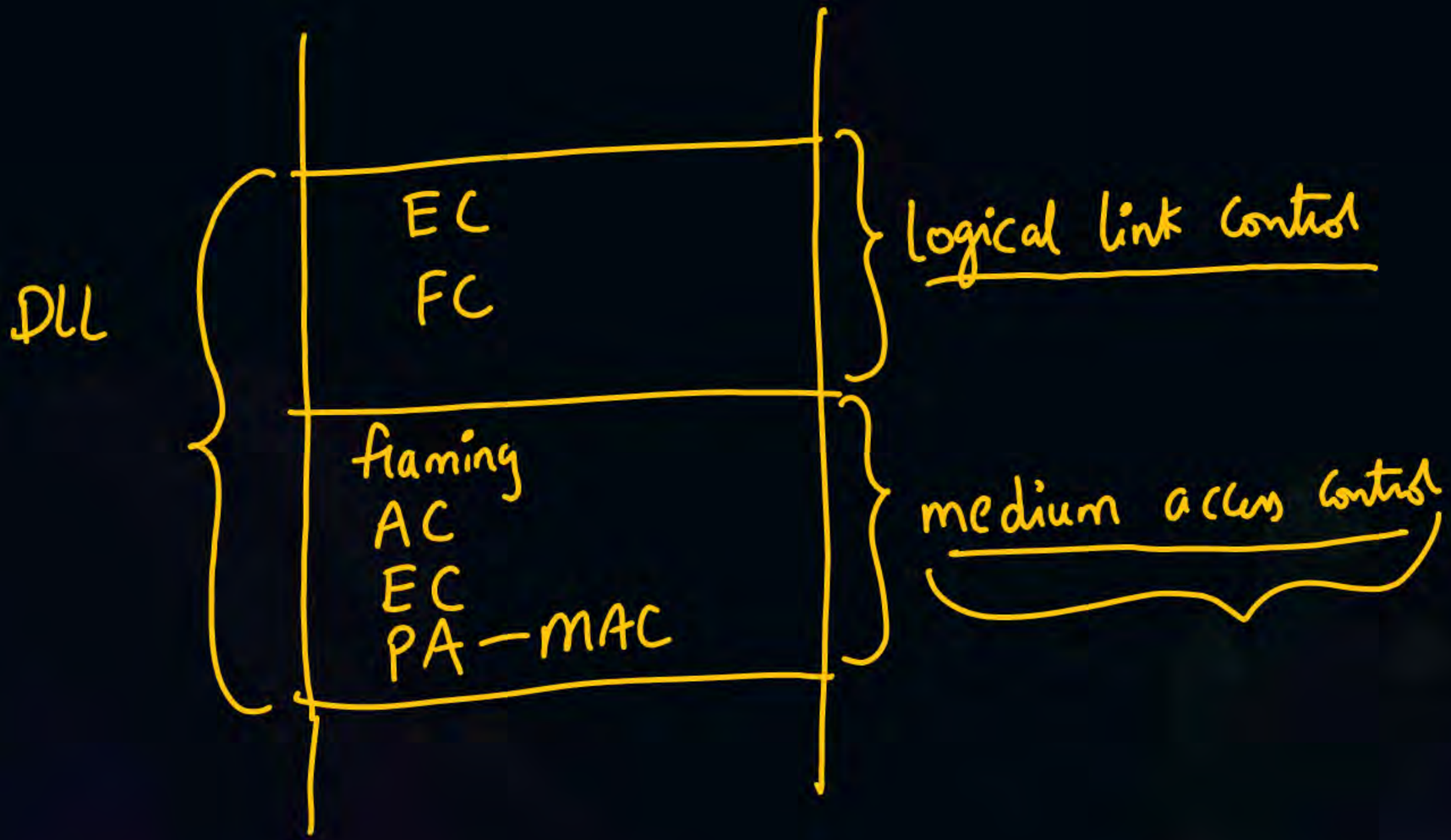
A) Yes, and we are using it. Because it is permanent

Ex:- Apple talk uses Random number as PA

Ethernet uses MAC as PA

MAC

Medium access Control

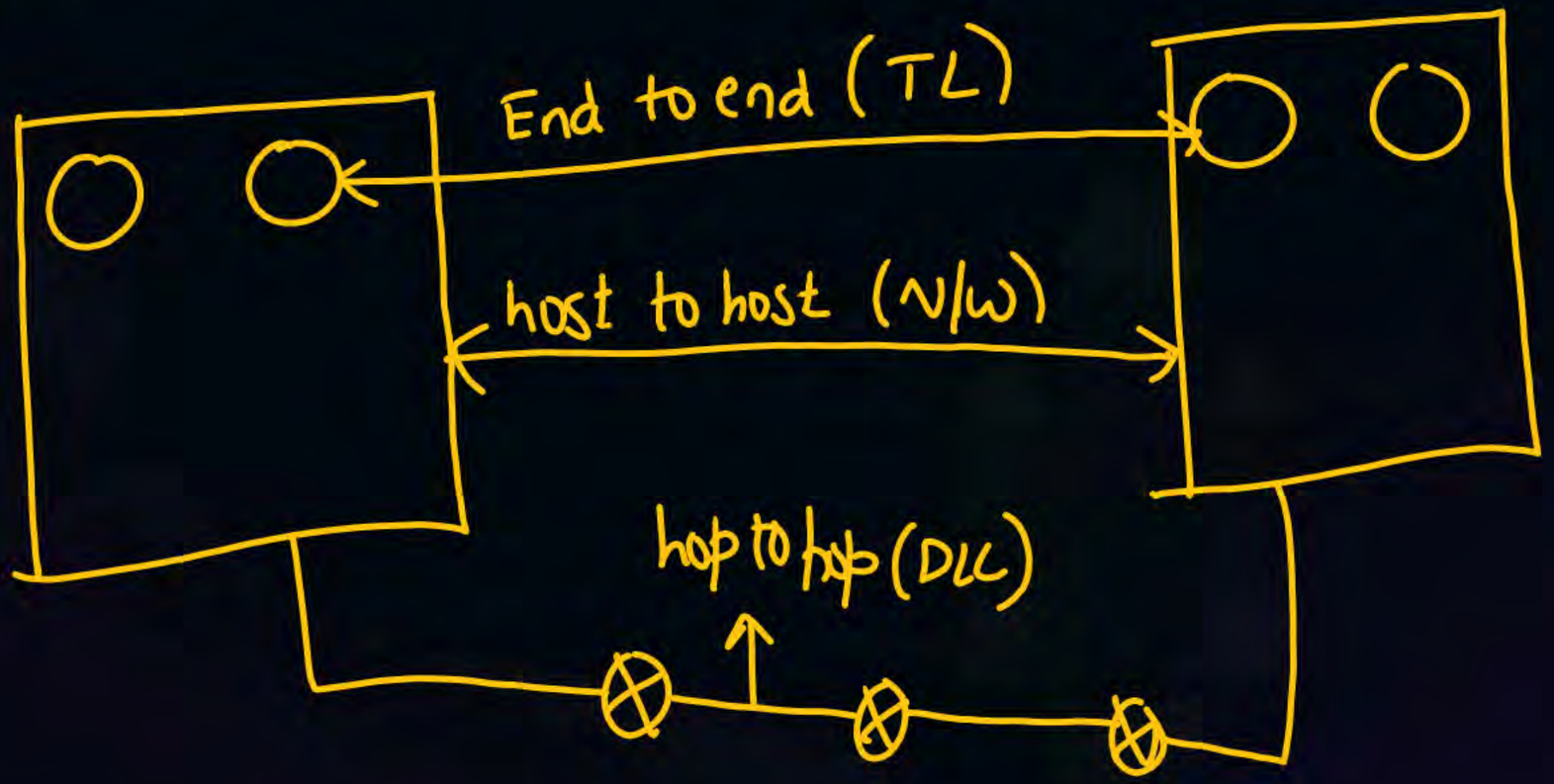

DLL
EC
FC
framing
AC
EC
PA-MAC
logical link control
medium access control


AL
PL
SL
TL
NL
DLL
PL

N/w layer :-

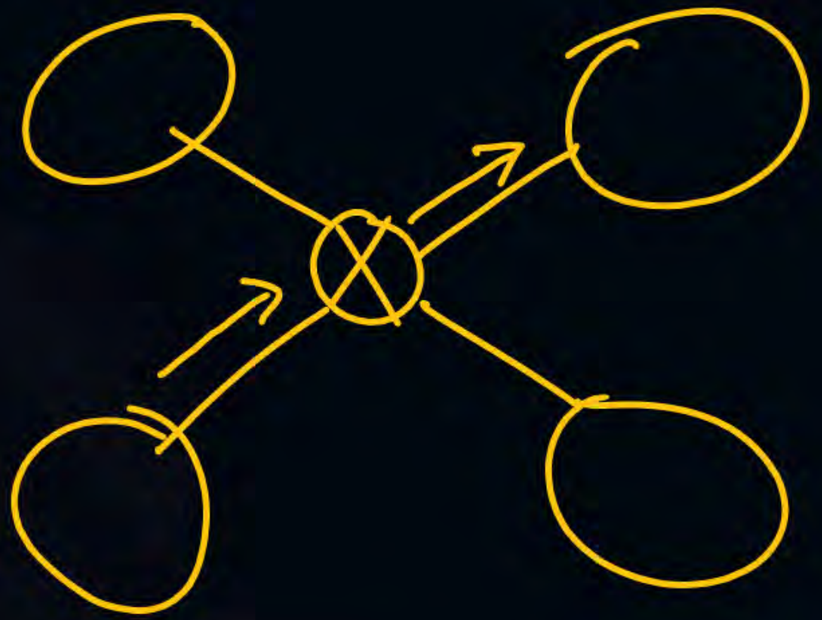
Responsibilities :-

1) Host to host Connectivity



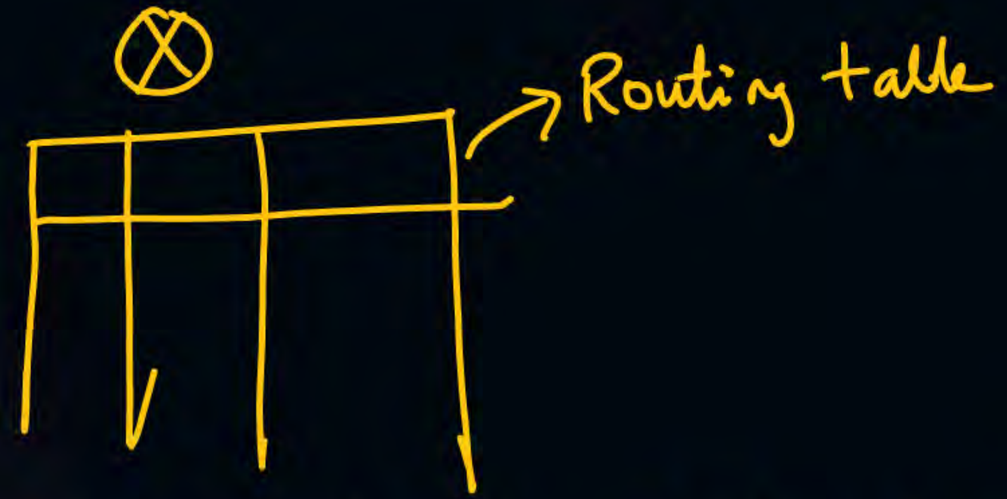
2) logical addressing — IP address

3) Switching



Switching :- taking an incoming packet and putting it on one of the outgoing links is called Switching.

(4) Routing :-

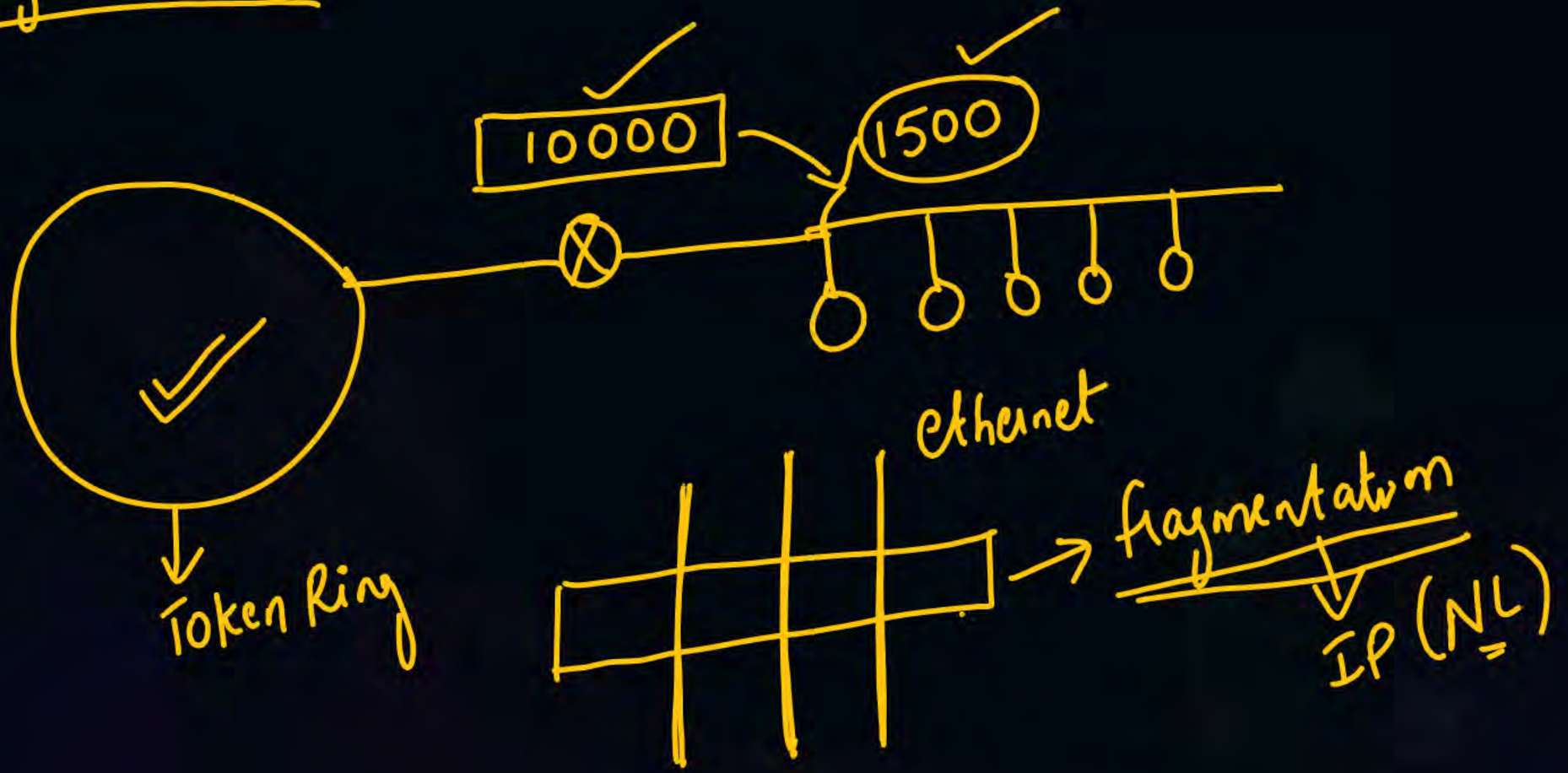


The process of building Routing table is called Routing

5) Congestion Control :- X

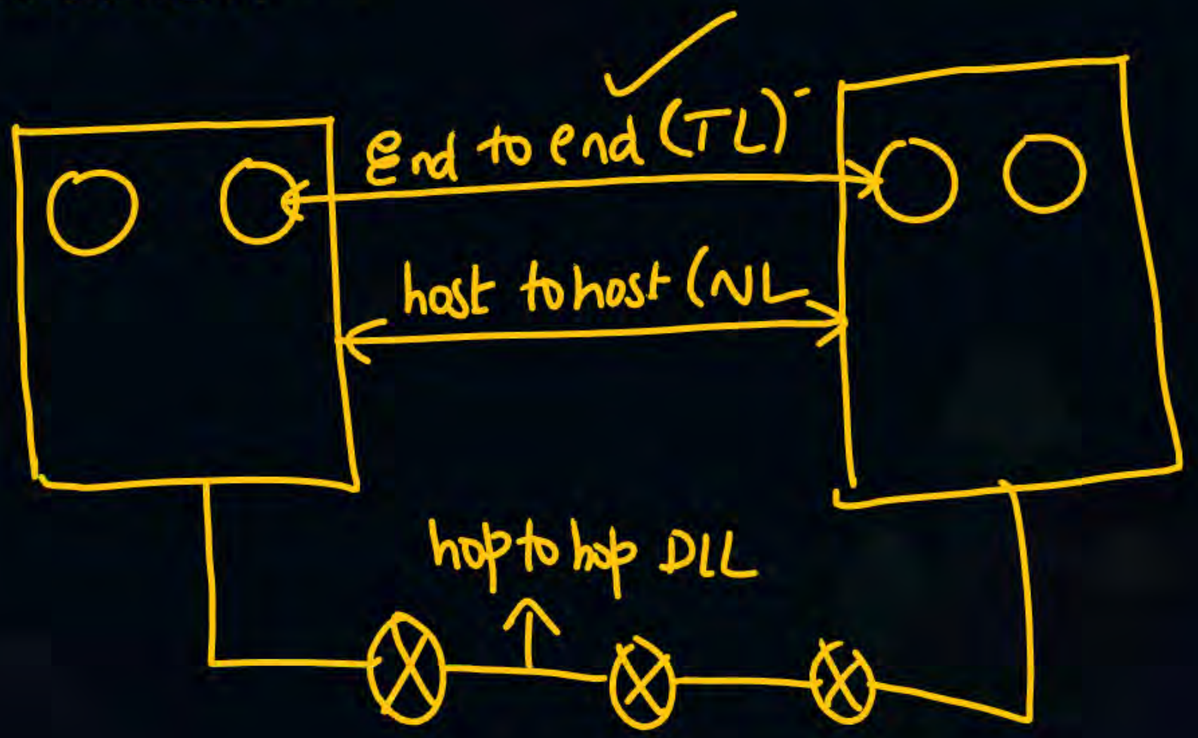
we have TCP/IP in Syllabus and IP doesn't do Congestion Control.

6) Fragmentation :-



Transport layer:-

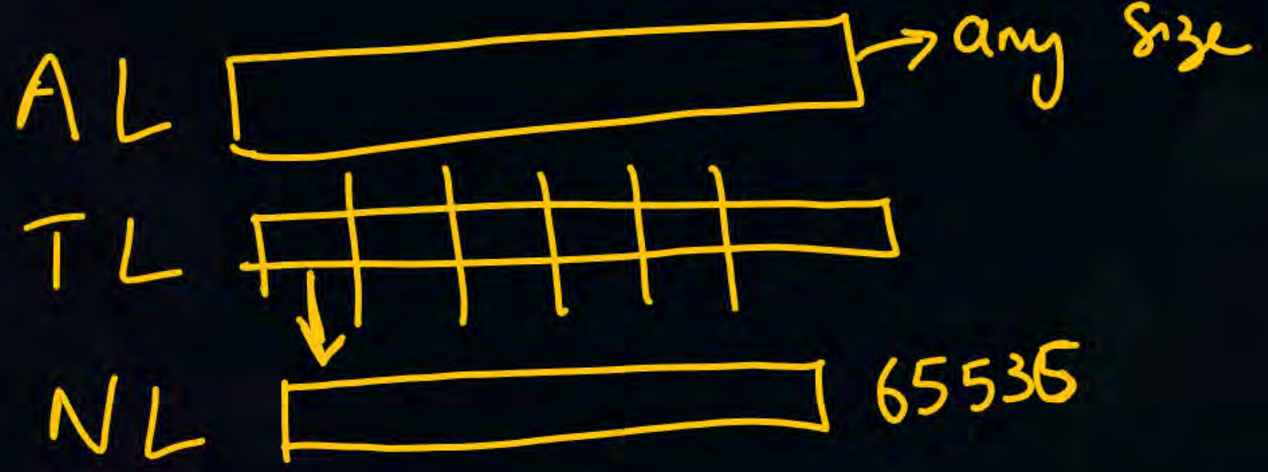
1) End to End Communication ✓



2) Flow control → SR

3) Error control (checksum)

4) Segmentation

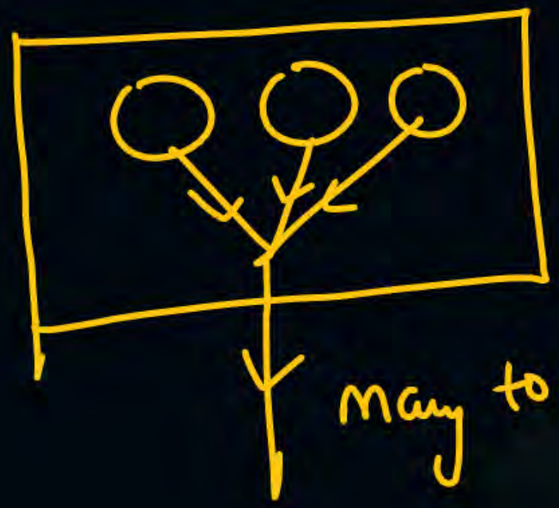


4) Segmentation :- AL can give data of any size to TL. TL will break it into segments so that they can fit in IP datagram this process is called segmentation. ✓

5) multiplexing and demultiplexing



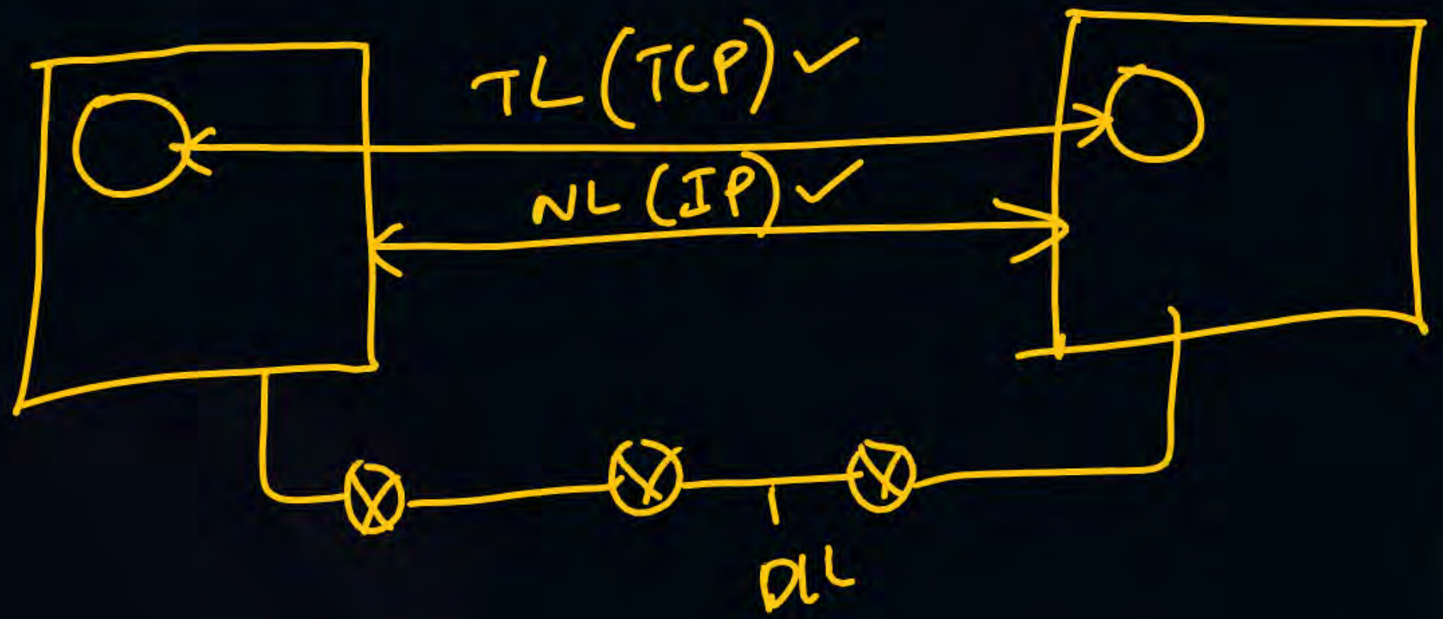
1 to many

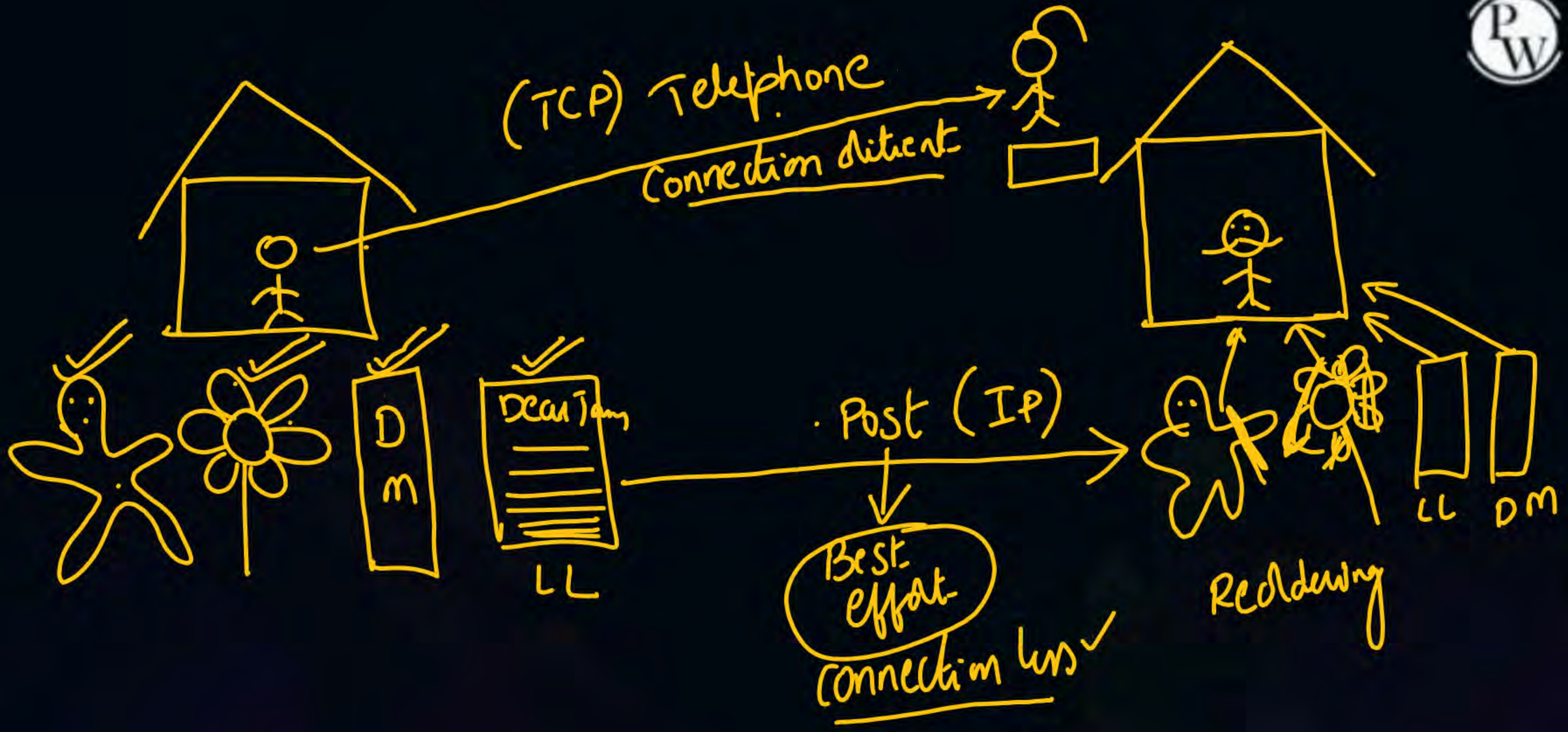


many to one

6) Congestion Control :- ✓

TCP at TL will do Congestion Control


TL (TCP) ✓
NL (IP) ✓
DLL


(TCP) Telephone
Connection oriented
DM
Dear Tanmay
LL
Post (IP)
Best effort
connectionless
LL DM
Reordering

THANK - YOU